

Group-Specific Employee Flow Rates

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Abstract

This paper extends the employee churn rate by introducing a group-specific measure of employee flows. The purpose of the measure is to permit analysis of changes in the composition of an employee population across identifiable employee groups. Rather than treating the workforce as a homogeneous population, the proposed framework separately records hiring and firing flows for each group. The resulting measure can be used to describe whether the representation of a group within an employee population is increasing, decreasing, or remaining stable over a specified interval. The measure is descriptive and does not, by itself, establish discriminatory intent or recruitment bias.

The paper ends with “The End”

Introduction

Many metrics are used in the field of human resources. The employee churn rate is one such metric. In the original formulation, the employee churn rate over an interval is determined by the difference between the number of employees hired and the number of employees fired, divided by the length of the observation interval.

This paper extends that framework to situations in which an employee population can be partitioned into identifiable groups.

Let the employee population be divided into groups indexed by $g \in \mathcal{G}$. Examples of groups may include organizational divisions, occupational categories, geographical offices, or other categories that are lawfully and appropriately available for analysis.

The objective is not to infer bias merely from differences in workforce composition. Instead, the objective is to provide a quantitative measure with which employee flows can be compared across groups.

The Employee Churn Rate

Consider two points in time t_1 and t_2 , where

$$t_2 > t_1.$$

The employee churn rate is given by

$$c(t_1, t_2) = \frac{h(t_1, t_2) - f(t_1, t_2)}{t_2 - t_1},$$

where

$c(t_1, t_2)$ is the employee churn rate from time t_1 to time t_2 ;

$h(t_1, t_2)$ is the number of employees hired from time t_1 to time t_2 ;

$f(t_1, t_2)$ is the number of employees fired from time t_1 to time t_2 .

The measure therefore describes the net employee flow per unit of time.

Partitioning the Employee Population

Suppose that the employee population is divided into G groups:

$$\mathcal{G} = \{1, 2, \dots, G\}.$$

For every group g , define

$$h_g(t_1, t_2)$$

as the number of employees in group g hired during the interval, and

$$f_g(t_1, t_2)$$

as the number of employees in group g fired during the interval.
The group-specific employee flow rate is then defined as

$$c_g(t_1, t_2) = \frac{h_g(t_1, t_2) - f_g(t_1, t_2)}{t_2 - t_1}.$$

This is the proposed measure.

Interpretation

The sign of $c_g(t_1, t_2)$ provides a simple description of the net direction of employee flow for group g .
If

$$c_g(t_1, t_2) > 0,$$

then the number of hires exceeds the number of firings for group g over the interval.
If

$$c_g(t_1, t_2) < 0,$$

then the number of firings exceeds the number of hires.
If

$$c_g(t_1, t_2) = 0,$$

then the number of hires and firings are equal.
The measure is therefore a flow measure rather than a direct measure of workforce representation.

Group Representation

To study workforce composition, define

$$N_g(t)$$

as the number of employees belonging to group g at time t , and define total employment as

$$N(t) = \sum_{g=1}^G N_g(t).$$

The representation of group g at time t is

$$p_g(t) = \frac{N_g(t)}{N(t)}.$$

Consequently,

$$\sum_{g=1}^G p_g(t) = 1.$$

The change in group representation between two points in time is

$$\Delta p_g = p_g(t_2) - p_g(t_1).$$

A positive value indicates that the group's share of total employment increased, while a negative value indicates that its share decreased.

The Group Representation Change Rate

For comparisons across intervals of different lengths, define the group representation change rate as

$$r_g(t_1, t_2) = \frac{p_g(t_2) - p_g(t_1)}{t_2 - t_1}.$$

This measure is distinct from the group-specific employee flow rate.

The first measure,

$$c_g(t_1, t_2),$$

measures the net flow of employees in absolute numbers per unit of time.

The second measure,

$$r_g(t_1, t_2),$$

measures the change in the group's share of total employment per unit of time.

The two measures should therefore not be treated as interchangeable.

Relative Group Flow

A further normalization can be obtained by dividing the net group flow by the group's initial employment.

For $N_g(t_1) > 0$, define

$$q_g(t_1, t_2) = \frac{h_g(t_1, t_2) - f_g(t_1, t_2)}{N_g(t_1)(t_2 - t_1)}.$$

The quantity q_g represents net employee flow relative to the group's initial size.

This normalization is useful when groups differ substantially in size.

For example, a net increase of ten employees has a very different meaning for a group initially containing one hundred employees than for a group initially containing ten thousand employees.

Comparing Two Groups

Consider two groups i and j . Their group-specific flow rates are

$$c_i(t_1, t_2) = \frac{h_i(t_1, t_2) - f_i(t_1, t_2)}{t_2 - t_1}$$

and

$$c_j(t_1, t_2) = \frac{h_j(t_1, t_2) - f_j(t_1, t_2)}{t_2 - t_1}.$$

Their difference is

$$D_{ij}(t_1, t_2) = c_i(t_1, t_2) - c_j(t_1, t_2).$$

Similarly, their normalized relative flows may be compared through

$$D_{ij}^q(t_1, t_2) = q_i(t_1, t_2) - q_j(t_1, t_2).$$

These quantities provide descriptive comparisons of employee flows.

Hiring and Firing Rates Separately

Because net flow can conceal offsetting gross flows, the framework should also retain the underlying hiring and firing rates.

Define

$$H_g(t_1, t_2) = \frac{h_g(t_1, t_2)}{t_2 - t_1}$$

and

$$F_g(t_1, t_2) = \frac{f_g(t_1, t_2)}{t_2 - t_1}.$$

Then

$$c_g(t_1, t_2) = H_g(t_1, t_2) - F_g(t_1, t_2).$$

This decomposition is important.

For example, two groups can have identical net employee flow while having very different hiring and firing rates.

Thus,

$$c_i = c_j$$

does not imply

$$H_i = H_j$$

or

$$F_i = F_j.$$

A Group Flow Vector

For G groups, define the group-flow vector

$$\mathbf{c}(t_1, t_2) = \begin{pmatrix} c_1(t_1, t_2) \\ c_2(t_1, t_2) \\ \vdots \\ c_G(t_1, t_2) \end{pmatrix}.$$

Similarly, define

$$\mathbf{H}(t_1, t_2) = \begin{pmatrix} H_1(t_1, t_2) \\ H_2(t_1, t_2) \\ \vdots \\ H_G(t_1, t_2) \end{pmatrix}$$

and

$$\mathbf{F}(t_1, t_2) = \begin{pmatrix} F_1(t_1, t_2) \\ F_2(t_1, t_2) \\ \vdots \\ F_G(t_1, t_2) \end{pmatrix}.$$

The fundamental relationship is

$$\mathbf{c} = \mathbf{H} - \mathbf{F}.$$

This representation permits the employee population to be analyzed as a multidimensional flow system.

A Composition-Adjusted Flow Measure

The representation of group g can change because of the group's own employee flows and because the total employee population changes.

A useful descriptive quantity is therefore

$$a_g(t_1, t_2) = \frac{c_g(t_1, t_2)}{N(t_1)}.$$

This expresses the group's net employee flow relative to total initial employment.

It should not be interpreted as a causal measure. It is a normalization that facilitates comparison between groups and organizations of different sizes.

Application to Workforce Composition

Suppose an organization contains several employee groups and wishes to study changes in workforce composition.

The following quantities can be calculated for each group:

Measure	Symbol	Interpretation
Hires	h_g	Number hired in group g
Firings	f_g	Number fired in group g
Net flow	c_g	Net employees per unit time
Hiring rate	H_g	Hires per unit time
Firing rate	F_g	Firings per unit time
Representation	p_g	Group share of employment
Representation change	Δp_g	Change in group share
Relative flow	q_g	Net flow relative to initial group size

Table 1: Group-specific employee-flow measures.

Together, these measures provide a more complete description of workforce dynamics than aggregate churn alone.

Bias and Causality

A difference in employee composition is not, by itself, evidence of recruitment bias.

For example, suppose

$$p_i(t) > p_j(t).$$

This establishes that group i has a greater representation than group j at time t . It does not establish why the difference exists.

Similarly,

$$c_i(t_1, t_2) > c_j(t_1, t_2)$$

establishes a difference in net employee flow, but does not establish that the difference resulted from discriminatory recruitment, retention, or termination practices.

A causal analysis would require additional information concerning the relevant employee population, applicant pools, qualifications, vacancies, selection procedures, compensation, promotions, turnover, and other potential explanatory variables.

Accordingly, the measures proposed in this paper should be understood as descriptive statistical instruments.

Data Requirements

A practical implementation requires, at minimum, observations of employee flows over a specified interval.

For each group g , the following quantities are required:

$$h_g(t_1, t_2), \quad f_g(t_1, t_2), \quad N_g(t_1), \quad N_g(t_2).$$

Total employment is also required:

$$N(t_1), \quad N(t_2).$$

The data should be collected and analyzed in accordance with applicable privacy, employment, and data-protection requirements.

Conclusion

The employee churn rate provides a measure of aggregate net employee flow. This paper extends that concept by defining the group-specific employee flow rate

$$c_g(t_1, t_2) = \frac{h_g(t_1, t_2) - f_g(t_1, t_2)}{t_2 - t_1}.$$

The framework additionally defines group representation,

$$p_g(t) = \frac{N_g(t)}{N(t)},$$

and representation change,

$$r_g(t_1, t_2) = \frac{p_g(t_2) - p_g(t_1)}{t_2 - t_1}.$$

These measures permit an organization to distinguish between employee flows and changes in workforce composition.

The principal advantage of the framework is therefore decomposition. Instead of observing only aggregate employee churn, an organization can observe how hiring, firing, net employee flows, and workforce representation evolve across groups.

The measures do not establish causality or discriminatory intent. Rather, they provide a quantitative foundation upon which subsequent statistical analysis can be constructed.

The End